

Deformation and Damage Evolution Characteristics in Laminated Rocks Using Discrete Element Method

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ABSTRACT: Laminated rocks, prevalent in natural geological processes, exhibit pronounced mechanical anisotropy, a characteristic shared with unconventional hydrocarbon reservoirs. The arrangement of laminations significantly influences fluid flow and stress propagation, underscoring the imperative to comprehend the mechanical behavior of such materials. Traditional homogenization models and continuum-based numerical methods often struggle to accurately capture the intricate dynamics of granular materials, marked by variations and arching mechanisms crucial for system stability. In contrast, discrete element methods (DEM) offer a distinct advantage by considering heterogeneity and physical interactions at the level of individual grains. This study evaluates force and stress evolution in composite laminated rocks, employing digital representations of laminated materials with varying volume fractions of binary mixtures. Utilizing a particle flow code, we construct digital rock samples consisting of rigid grain assemblage and a flat joint contact model governing grain-to-grain contacts.

Three digital rocks were first constructed: matrix, stiff material, and soft material. The stiff and soft materials are embedded into the matrix material, similar to laminated rock types. We then investigate the effect of relative lamina thickness and the angle of laminations with respect to loading directions. We observed that the number of laminations has little impact on the elastic stiffness and strength of the laminated rocks. The material exhibits higher effective stiffness when loaded parallel to the laminations, while stiffness is lower perpendicular to the laminations. The evolution of cracks proves intricate, reflecting a delicate interplay between material elastic constants, strength properties, and loading direction. We observed that the initiation and propagation of cracks show strong dependence on the angle of laminations and inclusion type. These findings contribute valuable insights into the mechanical behavior of laminated composite granular rocks.

1 INTRODUCTION

The effects of sedimentary processes, mineralogical differences and mechanical deformation result in the generation of various laminae within the Earth's crustal rocks. Anisotropic properties typically result from mineral alignment in metamorphic rocks, layering in sedimentary rocks, and discontinuities in the igneous rock mass (Chenevert and Gatlin, 1965; Williams, 1990). In geoengineering applications, the consideration of anisotropy is paramount. The geoengineering applications that encounter anisotropic rocks include drilling and hydraulic fracturing operations in thin-bedded, laminated shall sand hydrocarbon reservoirs, fractured basement rocks, shale layers, excavations, and geotechnical design of anisotropic rocks (Winhausen et al., 2023; Yan et al.; Ni et al., 2023).

Applications in engineering that overlook how rocks behave differently in various directions can lead to errors of varying degrees, depending on the extent of the rock's anisotropy Amadei (1996).

Numerous studies have explored the directional variations in rock strength and failure behaviors through Unconfined Compressive Strength (UCS), Brazilian tensile tests, and Triaxial tests, among others. Cho et al. (2012) conducted a systematic investigation into the anisotropic deformation and strength of igneous, metamorphic, and sedimentary rocks, measuring angles from 0 to 90 degrees in 15-degree increments relative to the transverse isotropic planes. Their findings revealed significant anisotropy, with up to a fourfold increase in Young's modulus and an eighteenfold increase in strength values. The analysis sug-

gested that failure modes are predominantly influenced by the orientation of weak layers relative to the loading direction.

Further research on various rock types has corroborated these results, underscoring the importance of considering anisotropic behavior in transversely isotropic rocks (Heng et al., 2020, 2015; Aliabadian et al., 2019). These studies highlight the complex mechanical behavior of laminated rocks but also acknowledge a gap in understanding the micromechanical behaviors during loading. An advancement in addressing this gap is the application of X-ray micro-Computed Tomography (micro-CT) imaging to study the damage behavior post-failure of artificial transverse isotropic rocks, as explored by Yang et al. (2019). This technique has provided significant insights into the locations and nature of microcracks, distinguishing between shear and tensile failure modes. However, it primarily captures conditions after failure, not during the onset of failure, where monitoring crack evolution could offer crucial insights into a rock's strength. The existing research highlights the limitations of traditional experimental methods in understanding the spatio-temporal evolution of cracks from the initial stages of loading. (Moradian et al., 2016).

The Discrete Element Method (DEM) is a numerical technique used in computational mechanics to model the behavior of granular materials and particulate systems. In DEM, the material is represented as a collection of discrete elements, each with its own properties, such as mass, shape, and interactions with neighboring elements. These elements interact through contact forces, simulating complex behaviors such as particle collisions, rolling, sliding, and compaction (CUNDALL and HART, 1993; O'Sullivan, 2011a). DEM stands out in creating intricate geometric arrangements of mineral aggregates without requiring homogenization. Importantly, it provides a distinct benefit by examining the distribution of force chains in granular rocks, the initiation and progression of cracks, pre-existing micro-crack deformations/slip and grain-scale deformations or dislocations. Analyzing these phenomena proves to be difficult with continuum-based approaches. Researchers have highlighted the benefits of employing the discrete element method to analyze the mechanical characteristics and micromechanical damage behavior of granular rocks and soils (Garcia and Bray, 2018; Singer et al.; Munjiza et al., 2020).

Previous studies used DEM that explicitly incorporates weak layers commonly found in naturally anisotropic laminated rocks to model these materials' strength and deformation behavior (Duan et al., 2016; Shi and Mishra, 2021;

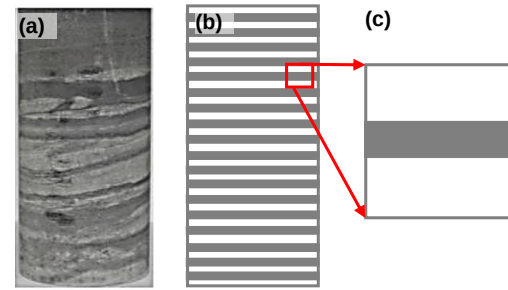


Fig. 1: Figure depicting laminated rock types: (a) Actual example of natural laminated rock from (Xu et al., 2023), (b) Schematic representation laminated rock, and (c) Representation of a single lamina used as an analogy in the current study.

Russo et al., 2017; Noori et al., 2024). This was achieved by introducing smooth joint contacts into the bonded particle model. Our study differs from the previous research in two ways; firstly, we examined the incorporation of thin layers instead of replacing the contacts with the smooth joint model. Secondly, the present study focuses on understanding the deformation and crack evaluation at the layer level of the laminated rocks rather than the macroscopic laminated rock, intending to understand the deformation and crack evolution of laminated rocks at the layer level (See Figure 1).

The primary objective of the present study is to investigate the directional variations in the deformation behavior and damage behavior of laminated rocks at the layer scale. Our analysis focuses on sedimentary rock types, where stratification is critical for anisotropy, such as in thin-bed reservoirs or laminated shaly sands Aldred (2021). Sandstone-shale layers of moderate strength typically form these reservoir rocks.

2 METHODOLOGY

We used Two Dimensional Particle Flow Code (PFC2D) for discrete element modeling (PFC7.0 Itasca Consulting Group, Inc, 2021). PFC2D consists of basic components such as disc-shaped rigid particles, walls, and interaction laws. These particles are characterized by specific size, rigid, and mass. Particles can undergo translational and rotational motion governed by Newton's force-displacement law. To start modeling with PFC2D, the initial phase is to create a packing of grains of desired distribution and properties (Figure 3), identify contacts and assign contact laws, define boundary conditions, and change boundary conditions to update the model state.

2.1. Packing

In the 2D modeling approach, each grain is depicted as a disc with a specified radius and unit thickness. The digi-

tal representation of a rock specimen takes the form of a square with dimensions of 50.8 mm by 50.8 mm. Four boundary walls or facets are constructed precisely aligned with the square's edges. Enclosed within this geometrical space is a random arrangement of grains, which are uniformly distributed across a defined radius range of 0.2 to 0.4 mm. The procedure for packing involves the following stages. Initially, parameters such as the boundaries, target porosity, grain radius range, and radius distribution are specified. Subsequently, grains are selected from the defined radius distribution until the aggregate solid fraction aligns with the desired porosity. These grains are then randomly positioned within the specified domain, leading to overlaps among grains. Such overlaps result in force imbalances and elevated kinetic energy levels. A cyclic process is then applied to bring the grain assembly into equilibrium, aiming to minimize overlaps and dissipate kinetic energy. Following this, displacements and net forces are nullified, establishing the model in its initial state. The decision to use a square specimen for the anisotropic properties investigation is driven by the ability to test it in different directions without changing the packing configuration.

2.2. Contact Law

The PFC calculates the interactions between particles at each timestep using Newton's second law of motion and the force-displacement principle. These interactions, which encompass forces and moments, are dynamically adjusted according to a predefined rule known as the contact law. In this research, the flat-joint model (FJM) was selected as the primary contact law, an advancement over the bonded particle model (BPM), with details provided in the P.F.C. manual and reiterated here the key ideas of the model (Itasca Consulting Group, 2023).

Contacts are initially established by identifying pairs of particles where the distance between them falls below a set threshold, termed the installation gap (g_0) (see Figure 2). Then, the FJM is assigned to these contacts. In the 2D framework, this interface is depicted as having a finite length and is segmented into a specified number (N) of distinct elements (see Figure 2). Each element is attributed with specific tensile and shear strengths, making them susceptible to fail under forces that exceed these thresholds. At each timestep, the forces and moments acting on each element are calculated and updated accordingly. The formula for the normal stress experienced by a bonded element within the flat joint model is given as (Itasca Consulting Group, 2023)

$$\sigma_N = k_N g, \quad (1)$$

where k_N represents the normal stiffness of the contact,

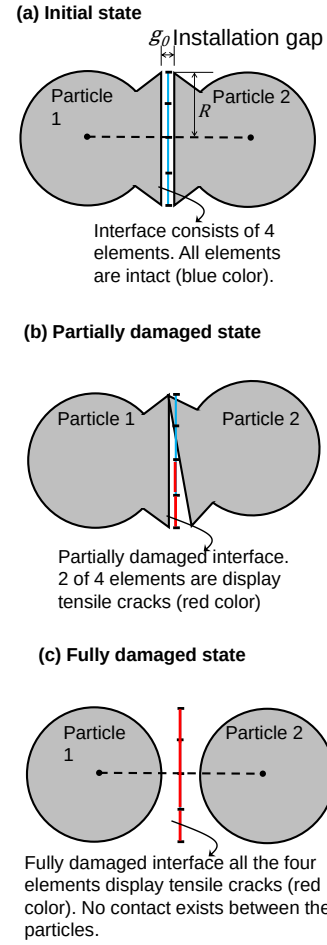


Fig. 2: Diagrammatic Representation of the Flat-Joint Model (FJM) Mechanics. (a) Initial Configuration: Establishment of contacts with designated contact properties and discretization of the contact interface into various elements. (b) Intermediate Damage Phase: Enables translational and rotational movement of particles. Interface elements, based on exerted forces, may attach or detach, leading to a partially bonded interface. (c) Complete Damage Stage: The interface undergoes frictional sliding or total detachment once all elements detach (Modified after (Pothana et al., 2024))

and g denotes the interface gap. Here, $\sigma_N > 0$ for $g > 0$ indicates tensile stress, whereas $\sigma_N < 0$ for $g < 0$ implies compressional stress. The calculation of the maximum shear stress, τ^{max} , follows from the relative displacement of each element, expressed as

$$\tau^{max} = k_S \delta U_s, \quad (2)$$

where k_S is the shear stiffness, and δU_s is the relative shear displacement. For a fully bonded element, the shear strength, τ_c , is formulated as

$$\tau_c = C + \sigma_N \tan(\phi), \quad (3)$$

with C being cohesion and ϕ the friction angle. In contrast, the shear strength in an unbonded frictional state is

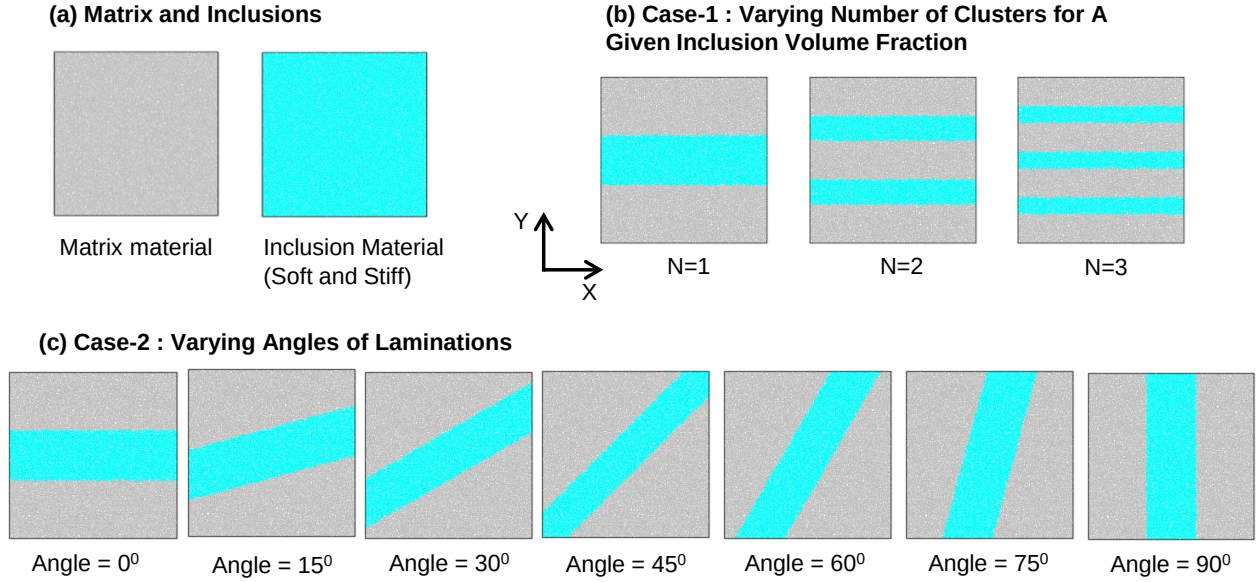


Fig. 3: Schematic of Packing and Geometrical Configurations in Current Research. (a) Initial Setup: Uniform material packing. (b) Case-1: Maintaining a constant volume fraction with an increase in the number of laminations. (c) Variation in Lamination Angle: Holding the volume fraction steady, the orientation of laminations relative to the X-axis is altered through a range from 0 to 90 degrees, in 15-degree increments.

defined as

$$\tau_r = -\mu\sigma_N, \quad (4)$$

where $\mu = \tan^{-1} \phi_r$ denotes the coefficient of friction, and ϕ_r represents the residual friction angle. Integrating the force and moment components across each interface element is essential for updating the particle velocity, adhering to Newton's force-displacement law.

Our study incorporates the deformability method from PFC to determine the values of k_N and k_S , utilizing the effective modulus (E^*) and k_{ratio} to specify these values indirectly. This approach allows the particle assembly's Young's modulus and Poisson's ratio to emerge as system properties. The relation between the k_N and k_S of particles with radii R_1 and R_2 is formulated in terms of E and k_{ratio} , providing a foundation for understanding the mechanical behavior of the interface.

The interface itself, comprising multiple elements, can exhibit a partially damaged state, indicative of a combination of intact and fractured elements. This characteristic of the FJM model facilitates the analysis of gradual contact failure between grains. Complete fracturing of all elements transitions the interface to a frictional state, enabling frictional sliding that limits particle rotation, thus still preserving the interlocking behavior of grains and reproducing a realistic ratio of compressional-to-tensile strength akin to actual hard rocks. The bond is broken when the difference in distance from one another particles surpasses

the flat joint's diameter, eliminating the contact. Subsequent interactions are governed by a linear contact model, simulating the interaction akin to that between two flat discs, marking a transition to a conventional linear model without the presence of a flat-joint interface.

Ensuring an optimal timestep is crucial for updating the model state and integrating Newton's laws numerically. A higher timestep can lead to system instability, causing undesired large displacements of particles. Conversely, a smaller timestep results in prolonged computational times. In the Particle Flow Code (PFC), an automatic timestep determination algorithm is employed to establish a critical timestep based on stiffness and displacement constraints, as detailed in Itasca Consulting Group. The actual timestep used is a fraction of this critical timestep. The critical timestep, in turn, is influenced by factors such as contact stiffness, mass (density), the number of contacts, and the contact law. For an in-depth exploration of timestep determination, refer to O'Sullivan (2011b).

2.3. Model Construction

2.3.1 Geometrical Configuration of Inclusions

The mechanical properties of binary mixtures are being studied using three specific types of materials (see Figure 3). The first type, known as the matrix, is similar to a moderate-strength sandstone rock. The second type, called stiff inclusions, comprises hard mineral aggregates. The third type, termed soft inclusions, comprises soft mineral aggregates. The properties of the moderate strength

sandstone are based on the work of (Deere and Miller, 1966), while the properties of the stiff and soft inclusions are selected relative to the matrix material.

In this study, inclusions are systematically integrated into the matrix material, following a sequence of increasing volume fractions ranging from 0.1 to 0.5 in increments of 0.1 (v/v). For each volume fraction, the region corresponding to the inclusion is identified within the matrix, and the contact properties within this region are replaced with the contact properties of inclusions (see Figure 3). It is important to note that the porosity and grain size remain constant for both the inclusions and the matrix. The variation in properties is attributed to the differences in contact properties and grain properties within the designated inclusion area.

The analysis is bifurcated into two distinct cases: i) Variation in the number of layers for each volume fraction and ii) Alteration of the angle of laminations relative to the loading direction (see Figure 3). In the first case, the number of laminations is increased for a given volume fraction of inclusion, ensuring the cumulative volume fraction remains constant. This is equivalent to a decrease in the relative thickness of the laminations. In the second case, for a specific volume fraction (0.3 in this study), lamination angles vary from 0 to 90 degrees in increments of 15 degrees relative to the horizontal axis or the transverse isotropic plane. Figure 3 illustrates the geometrical configurations for each case.

2.4. Numerical Simulation

We performed a strain-controlled Unconfined Compression Test (UCS) on the digital rock specimens. The UCS is a widely used rock test that allows us to estimate critical properties such as Young's modulus, Poisson's ratio, and the rock's failure characteristics under axial loading. Here's a breakdown of the two cases: In Case-1 (as depicted in Figure 3)), we apply an axial load separately along both the X and Y directions. In Case 2, the axial load is applied only along the Y-direction. Details regarding the UCS testing process are provided in the following description.

Initially, the side walls aligned with the loading direction are removed, and frictionless conditions are set for grain-wall contacts. Subsequently, a velocity of 0.005 m/s is applied in opposite directions to both loading walls, causing compression in the sample as illustrated in Figure 4. Throughout each cycle, the simulation monitors and records axial stress, axial strain, radial strain, and the bond-state of each contact.

- To determine axial forces, the combined force ex-

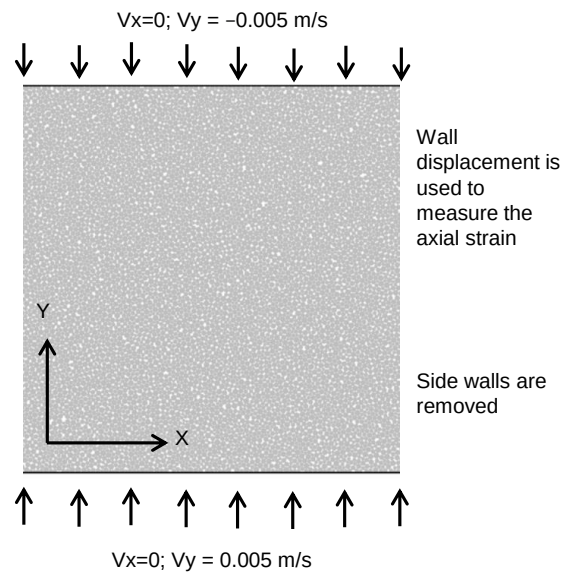


Fig. 4: Schematic diagram of the unconfined compression test used in the present study

erted on each wall in the direction of the load is calculated. Axial stress is then obtained by dividing the mean of these forces by the initial cross-sectional area of the specimen.

- Axial strain is calculated by dividing the total displacement of walls along the loading direction by the initial sample length.
- For radial strain calculation, the average lateral displacement of particles located on opposite sides and perpendicular to the loading direction of the specimen is considered. It is to be noted that lateral deformation may not be uniform along the sample length. This could affect the accuracy of the average lateral displacement as an estimate.
- The contact bond-state is a measure used to determine whether a bond is intact or broken. In the event of a breakage, the nature of the failure, whether it is due to tension or shear, is recorded. Additionally, the position and direction of the contacts at the point of failure are recorded.

The simulation concludes when the post-failure axial stress drops below half of the peak axial stress value. Young's modulus (E) is calculated by determining the slope of the regression line connecting axial stress and axial strain values, while Poisson's ratio (ν) is determined from the slope of the regression line between radial strain and axial strain values.

Table 1: PFC micro parameters

Parameter	Matrix	Soft Inclusion	Stiff Inclusion	Units
Particle Density	2600.0	2500.0	2650.0	kg/m^3
FJB* Effective modulus	50.0	10.0	90.0	GPa
FJB Normal-to-shear stiffness ratio	2.5	5.0	2.0	
FJB Tensile strength	15.0	10.0	20.0	MPa
FJB cohesion	10.0	6.0	15.0	MPa
FJB Friction angle	5.0	0.5	10.0	deg
FJB Surface installation gap	0.02	0.02	0.02	mm
Damping Coefficient	0.30	0.30	0.30	

* Flat Joint Bond

Table 2: Mechanical and strength properties of stiff material, matrix material and soft material

Mat.	Dir.	E (GPa)		PR		UCS(MPa)	
		mean	std	mean	std	mean	std
Stiff	x	93.06	0.20	0.183	0.002	34.54	0.31
Stiff	y	93.21	0.41	0.182	0.001	33.97	0.94
Matrix	x	49.16	0.13	0.226	0.003	21.85	0.55
Matrix	y	49.25	0.25	0.225	0.002	21.09	0.37
Soft	x	8.61	0.04	0.333	0.004	13.95	0.57
Soft	y	8.63	0.06	0.333	0.004	13.35	0.38

3 RESULTS AND DISCUSSION

3.1. Models and Calibration

In the preceding section, we constructed three distinct sets of digital rocks by adjusting microscopic parameters to attain the desired macroscopic behavior. The adjusted PFC2D micro parameters are tabulated in the Table 1. Figure 5 depicts the stress-strain response observed during axial loading. Table 2 provides the two-dimensional Young's modulus, Poisson's ratio, and UCS values for the matrix material, stiff inclusions, and soft inclusions. A comparative analysis is presented in Figure 6. The matrix material demonstrates a Young's modulus (E) of approximately 49.2 GPa, Poisson's ratio (ν) of approximately 0.225, and UCS value of approximately 21.4 MPa. These matrix properties position themselves between those of the stiff and soft materials. Stiff material exhibits the highest values of E and UCS while having the lowest ν , whereas soft material is characterized by the lowest E and UCS with the highest ν (refer to Figure 6).

The stress-strain curve responses of homogeneous materials, as depicted in Figure 5, reveal a notable similarity in behavior along both the X and Y directions. Furthermore, Figure 6 illustrates that the values of E, ν , and UCS are equivalent in both directions. This alignment is particularly advantageous for our current investigation, as the embedding of inclusions in rock, such as laminations, introduces macroscopic anisotropy. To rigorously assess this anisotropy, it is imperative to ensure that the initial material possesses isotropic characteristics. Any subsequent emergence of anisotropy now can be attributed to

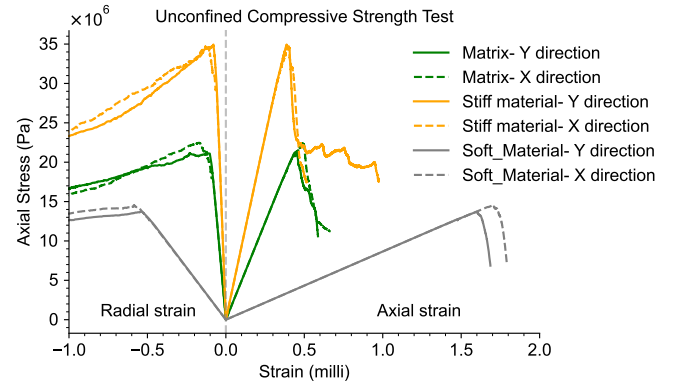


Fig. 5: The stress-strain response curves for three materials were examined in both X and Y directions. The close match in stress-strain behavior along both axes highlights the isotropic nature of the specimen.

the presence of these inclusions.

3.2. Effect of Inclusion Volume Fraction and Number of Laminations

The volume fractions of the inclusions are varied from 0.1 to 0.5 in increments of 0.1. For each specified volume fraction, laminations were systematically increased from 1 to 3, as illustrated in Figure 3, encompassing both stiff and soft inclusions. This can be seen as a decrease in relative lamina thickness with respect to the matrix volume or the sample dimensions.

The ensuing outcomes are graphically depicted in Figures 7, 9, and 10. These figures illustrate a clear trend: Young's modulus (E), Poisson's ratio (ν), and UCS rise as the volume of stiff material increases and decline as the volume of soft material increases. The findings presented align with the experimental studies conducted by Li et al. (2021), where they incorporated relatively soft inclusions (Simulated shale and sandstone) within a rigid matrix (Simulated Limestone). Their research demonstrates a increase in Young's modulus and UCS as the inclusion volume decreases, as illustrated in Figure 8.

It is evident that the number of laminations (or relative

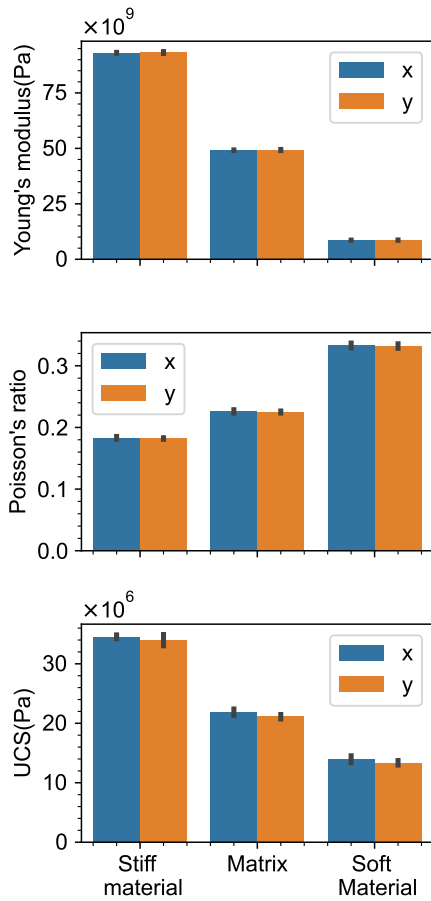


Fig. 6: Two-dimensional elastic and strength properties comparison among three homogeneous materials, featuring (a) Young's Modulus (E), (b) Poisson's Ratio (ν), and (c) Uniaxial Compressive Strength (UCS). The Error line on top of each bar represents the standard deviation observed during different random realizations of the packing.

lamination's thickness) has a negligible impact on the effective Young's modulus. Similarly, the Poisson's ratio (ν) remains invariant to the number of laminations when subjected to loading along the X direction (parallel to laminations). Conversely, significant fluctuations in ν manifest when the loading is oriented along the Y direction (perpendicular to laminations), as observed in Figure 9.

It is noteworthy that Poisson's ratio computation involves the utilization of radial strain, determined by the displacement of particles situated at the boundary, rather than relying on the overall average across the sample. The observed non-uniform radial displacement in both matrix and inclusion materials may contribute to the observed variations in ν during Y-direction loading. Nevertheless, a comprehensive investigation is necessary to understand this phenomenon in terms of spatial strain distribution. This would also provide insights the implications for rock mechanical tests conducted on laminated rocks. Such investigations

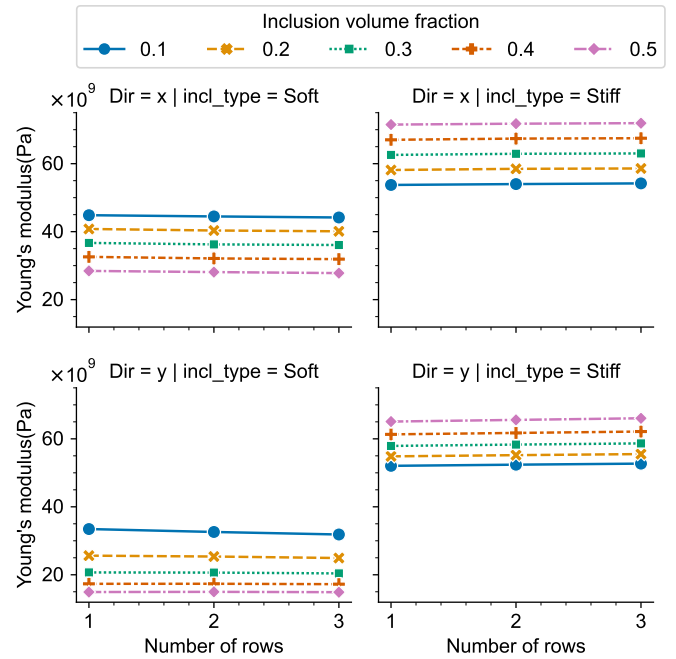


Fig. 7: Impact of Volume Fraction and Lamination Count on the Two-Dimensional Young's Modulus. The four subplots illustrate distinct scenarios corresponding to inclusion types (soft/stiff) and loading directions, revealing subtle variations in material behavior.

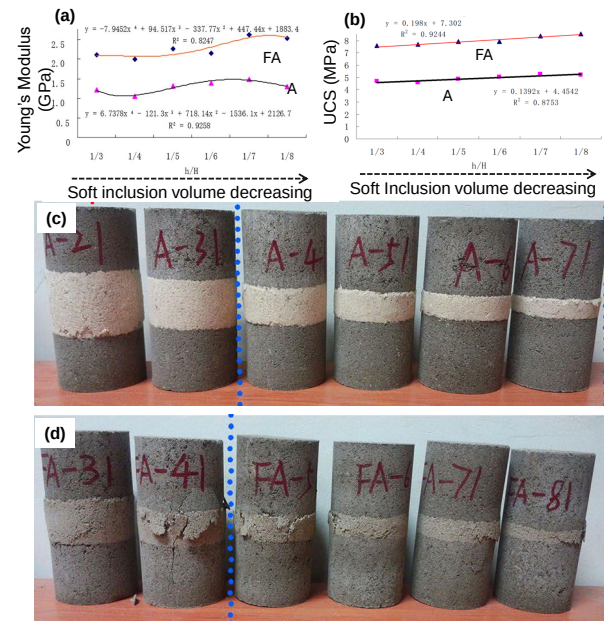


Fig. 8: Experimental investigation of laminated composites conducted by Li et al. (2021), depicting: (a) the variation of effective Young's modulus in relation to the inclusion volume, and (b) the Unconfined Compressive Strength (UCS) to the inclusion volume fraction. (c) Shale is sandwiched between limestone (d) Sandstone is sandwiched between limestone (Modified after Li et al. (2021))

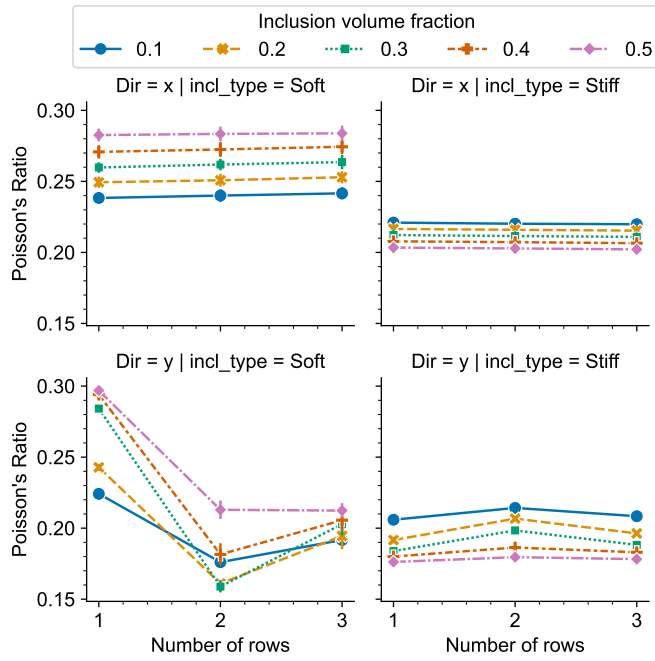


Fig. 9: Impact of Volume Fraction and Lamination Count on the Two-Dimensional Poisson's ratio. The four subplots illustrate distinct scenarios corresponding to inclusion types (soft/stiff) and loading directions, revealing subtle variations in material behavior,

will enhance the understanding of the impact of laminations on radial displacements, particularly when measured using strain gauges or LVDTs mounted to the outer surface of the sample.

Similarly, the quantity of laminations demonstrates minimal influence on the UCS value of the sample (refer to Figure 10). Nevertheless, it is noteworthy that the standard deviation is comparatively higher in the case of UCS when compared to the values of Young's Modulus (E) and Poisson's ratio (ν) across various random realizations.

The evolution of cracks during the Unconfined Compressive Strength (UCS) test, conducted with loading perpendicular to the laminations (as depicted in Figure 12) and parallel to the laminations (as shown in Figure 13), is examined. When subjected to perpendicular loading, both the initial set of cracks and those preceding the sample's yield strength primarily occur within the matrix, irrespective of inclusion type—whether soft or stiff. After failure, cracks evolved into the inclusion region. No distinct pattern is observed in the crack evolution behavior between the rigid and soft inclusions.

However, a notable divergence between soft and stiff inclusions manifests when subjected to loading parallel to the laminations. In instances involving stiff laminations, a majority of cracks initiate and coalesce within

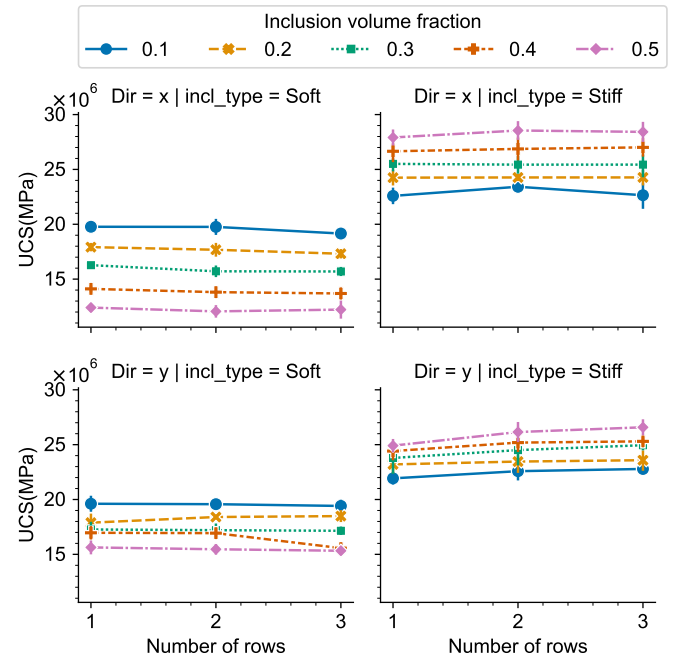


Fig. 10: Impact of Volume Fraction and Lamination Count on the Two-Dimensional UCS. The four subplots illustrate distinct scenarios corresponding to inclusion types (soft/stiff) and loading directions, revealing subtle variations in material behavior.

the stiff inclusion region. Conversely, in cases involving soft inclusions, the preponderance of cracks initiates and evolves within the matrix region, known for its relatively greater strength compared to soft inclusions. This divergence arises due to the stiffer material enduring heightened stresses, a consequence of near-iso-strain conditions experienced by the sample. Consequently, the analysis underscores that loading along the lamination direction exerts a more pronounced influence on crack evolution and propagation than loading perpendicular to the laminations. Furthermore, the number of laminations or the relative thickness of the laminations has little effect on the UCS value and crack evolution behavior.

3.3. Effect of Angle of Laminations

In this investigation, the influence of the lamination angle in relation to the loading direction is systematically examined. The loading direction is consistently maintained along the Y-axis, and the inclusion volume fraction remains a constant at 0.3 (v/v). Based on the preceding analysis, it was established that varying the number of laminations for a given volume (or relative lamina thickness) has a negligible impact on the elastic modulus (E), Poisson's ratio (ν), and unconfined compressive strength (UCS). Consequently, in this present analysis, the number of laminations remains unchanged, with the focus solely on utilizing single laminations.

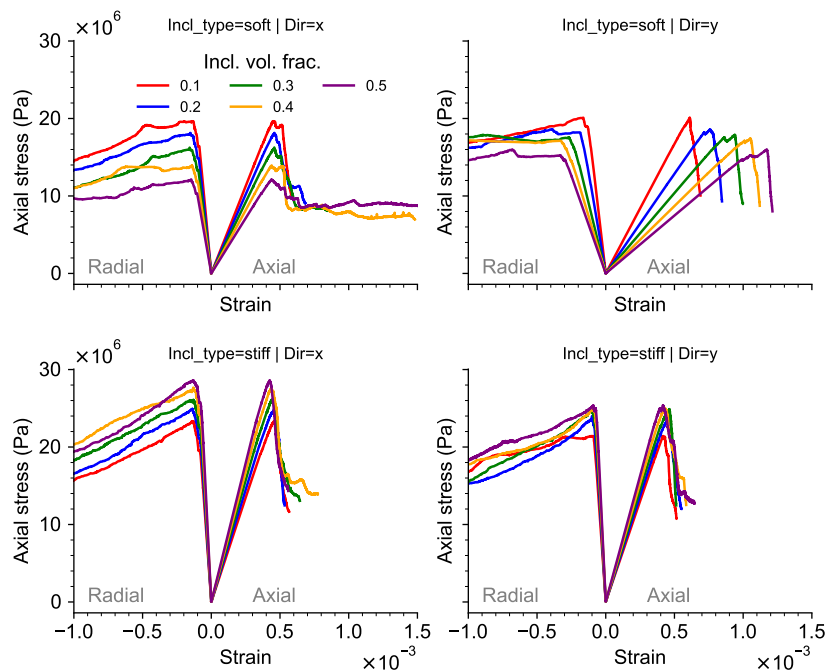


Fig. 11: Impact of volume fraction stress-strain response during the UCS test. The four subplots illustrate distinct scenarios corresponding to inclusion types (soft and stiff) and loading directions

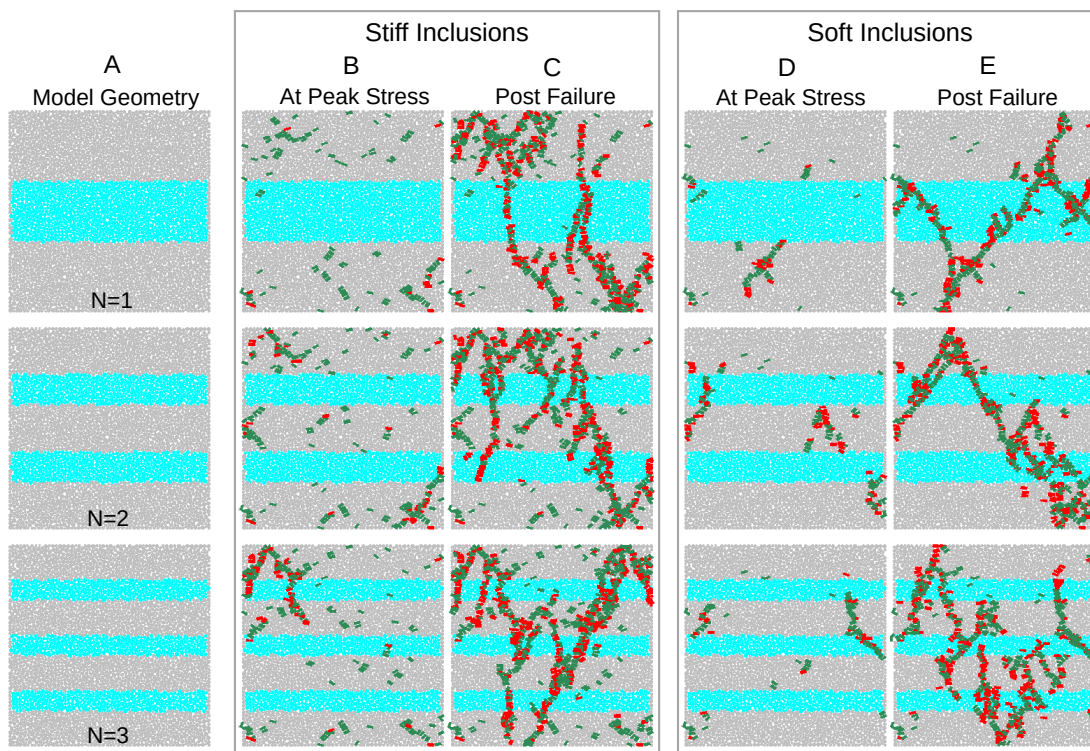


Fig. 12: Dynamic visualization of crack evolution during UCS test loading along Y-direction. Column (A) denotes model geometry (B) and (C) represent the crack distribution at the peak strength and post-failure of stiff inclusions. (D) and (E) show the crack distribution at the peak strength and post-failure stage. Green color and red color represent the shear and tensile cracks, respectively.

The measurement of lamination angles is conducted with respect to the X-axis in an anticlockwise direction, as il-

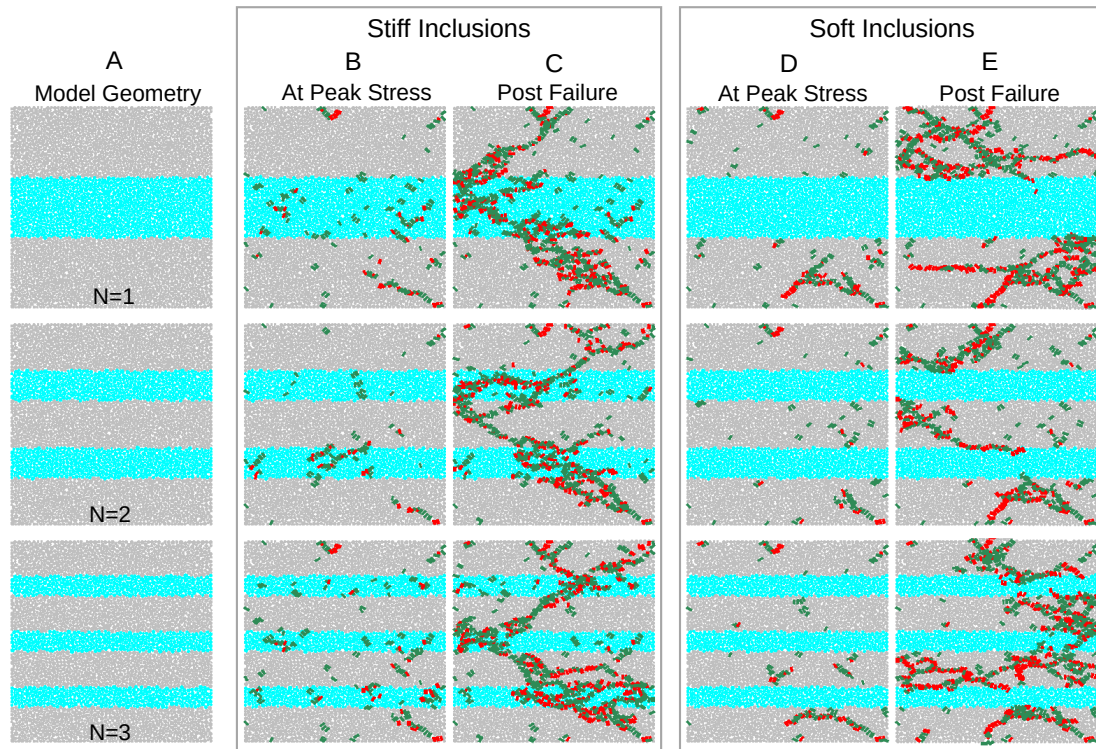


Fig. 13: Dynamic visualization of crack evolution during UCS test loading along X-direction. Column (A) denotes model geometry (B) and (C) represent the crack distribution at the peak strength and post-failure of stiff inclusions. (D) and (E) show the crack distribution at the peak strength and post-failure stage. Green color and red color represent the shear and tensile cracks, respectively.

illustrated in Figure 3. Sequentially, the angles are incremented from 0 to 90 degrees at intervals of 15 degrees, as depicted in Figure 3. Similar to the preceding cases, the particle arrangements are generated through three distinct random realizations. The obtained results are graphically presented with error bars, representing the standard deviation observed across different random realizations. Furthermore, this analysis distinguishes itself by considering both stiff and soft inclusions separately.

The stress-strain responses during UCS test are depicted in Figure 14. The graphical representation reveals a conspicuous influence of the lamination angle on the stress-strain evolution behavior, particularly notable during the presence of soft inclusions as opposed to stiff inclusions. Subsequently, the Young's modulus (E), Poisson's ratio (ν), and Unconfined Compressive Strength (UCS) are derived from the recorded stress-strain responses.

Figure 15 illustrates the relationship between Young's modulus and the angle of laminations. Notably, up to an angle of 45 degrees, Young's modulus remains relatively constant. Beyond this threshold, a discernible escalation in Young's modulus is observed, culminating in saturation at angles approaching 90 degrees. This consistent trend is evident for both soft and stiff inclusions, underscoring the uniform behavior of Young's modulus with

varying lamination angles. The findings align with the experimental results presented by Winhausen et al. (2023) on anisotropic shale rocks. In their study, it was observed that the Young's modulus exhibits a continuous and steady increase beyond angles of 45 degrees.

The figure presented in 16 illustrates the effective Poisson's ratio across distinct lamina angles. Unlike the linear behavior observed in Young's modulus, the progression of Poisson's ratio with respect to angle is intricate. Initially, there is an ascent in ν , followed by a subsequent descent. This deviation is more pronounced in instances featuring soft inclusions compared to those with stiff inclusions within the matrix.

It is imperative to acknowledge that the determination of ν relies on the displacement of particles confined to the boundary of the specimen. Consequently, it can be inferred that the radial displacement is comparatively diminished when the applied load is either parallel or perpendicular to the laminations. Notably, ν experiences decrease in value for angles intermediate to the aforementioned orientations.

In alignment with prior findings, it is observed that the Unconfined Compressive Strength (UCS) exhibits a greater standard deviation compared to E and ν across the three

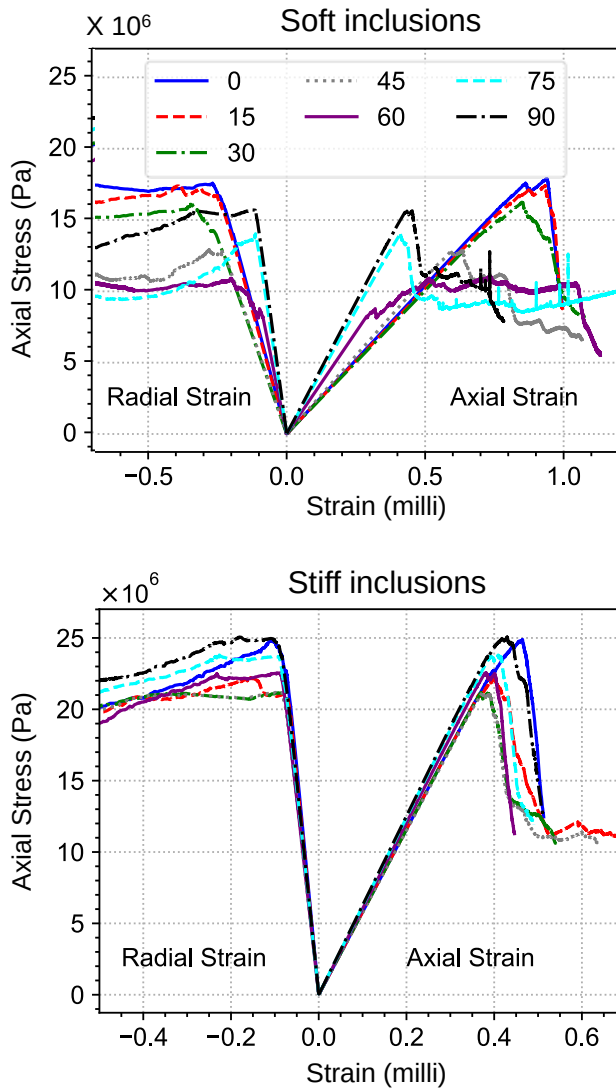


Fig. 14: Impact of the angle of laminations on the stress-strain response during the UCS test. Loading is along the Y direction

random realizations, as illustrated in Figure 17. Upon the incorporation of soft inclusions within the matrix, the UCS reaches its minimum at an inclination angle of approximately 60 degrees. The findings align with the experimental results presented by Winhausen et al. (2023) on anisotropic shale rocks. Their study observed that the UCS exhibits the lowest values when the angle of laminations is 45 to 60 degrees with respect to the transverse isotropic plane. Conversely, in the case of stiff inclusions, the UCS attains its lowest values within the range of 30 to 40 degrees. The findings align with the experimental results presented by Noori et al. (2024) for layered sandstone rocks. Their study revealed a similar trend between the UCS and the angle of anisotropy relative to the loading direction. Additionally, various UCS results compiled by Plinninger and Alber (2015) on metamorphic rocks show

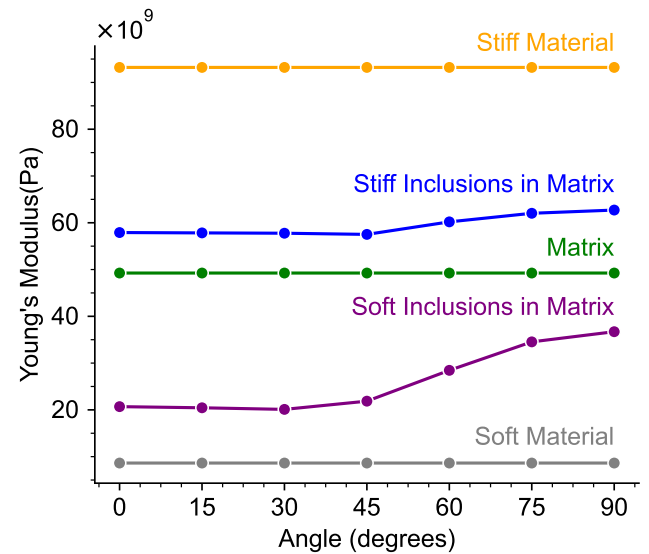


Fig. 15: Effect of angle of laminations with respect to the loading direction (Y-direction) on the two-dimensional Young's modulus for the soft and stiff inclusions in the matrix, along with the homogeneous materials. Vertical bars at each point represent the standard deviation observed during different random realizations of the packing.

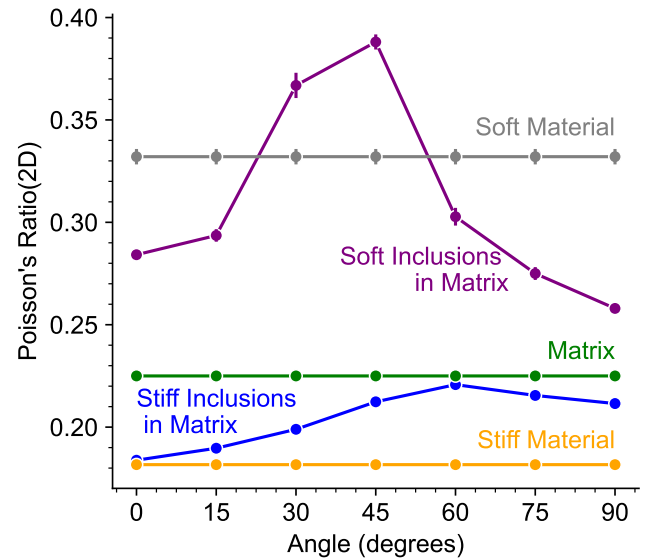


Fig. 16: Effect of angle of laminations with respect to the loading direction (Y-direction) on the two-dimensional Poisson's ratio for the soft and stiff inclusions in the matrix, along with the homogeneous materials. Vertical bars at each point represent the standard deviation observed during different random realizations of the packing.

a similar behaviour to that of the present findings. The implications of these outcomes suggest that the UCS values are influenced by the angle and positioning of the mate-

rial relative to the boundaries, irrespective of whether the inclusions possess high or low strength

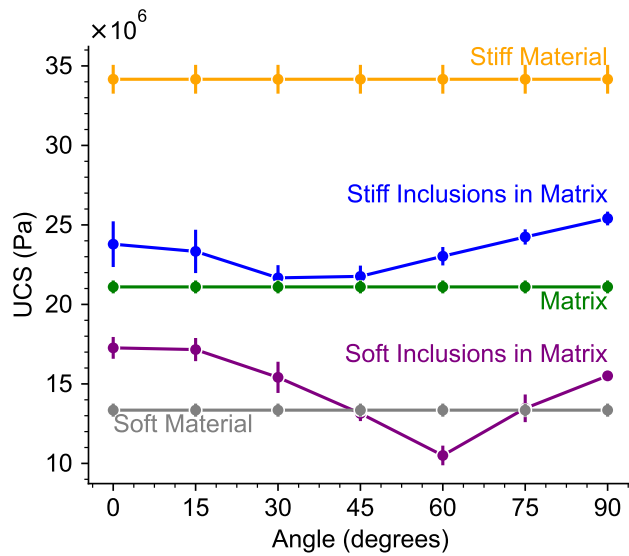


Fig. 17: Effect of angle of laminations with respect to the loading direction (Y-direction) on the two-dimensional UCS value for the soft and stiff inclusions in the matrix, along with the homogeneous materials. Vertical bars at each point represent the standard deviation observed during different random realizations of the packing.

The analysis of crack evolution was conducted under various scenarios. Figure 18 delineates the proportion of cracks localized within the matrix relative to the overall number of cracks at the point of peak strength attainment in the sample. The presented findings underscore the substantial influence of the lamination angle on crack distribution. Notably, when the angle of laminations falls below 45 degrees, cracks predominantly manifest in regions characterized by lower strength. Conversely, at angles exceeding 45 degrees, crack development is observed in areas of heightened strength due to the predominant load-bearing capacity of the stiff regions.

The graphical representation of tensile and shear crack propagation during both peak stress and post-failure stages is depicted in Figure 19. Notably, a discernible disparity in crack evolution is evident between soft and stiff inclusions. Specifically, at angles below 45 degrees, the preponderance of cracks manifests within the matrix for stiff inclusions and, conversely, within the inclusions for soft counterparts. Conversely, beyond the 45-degree threshold, a reversal in behavior is observed.

Of particular interest is the failure mechanism occurring at near-perpendicular angles (75 and 90 degrees), wherein the specimen experiences failure attributable to cracks

evolving predominantly within regions characterized by heightened stiffness, regardless of inclusion types. This outcome is ascribed by the previously explained concept of elevated stress concentration within these locations.

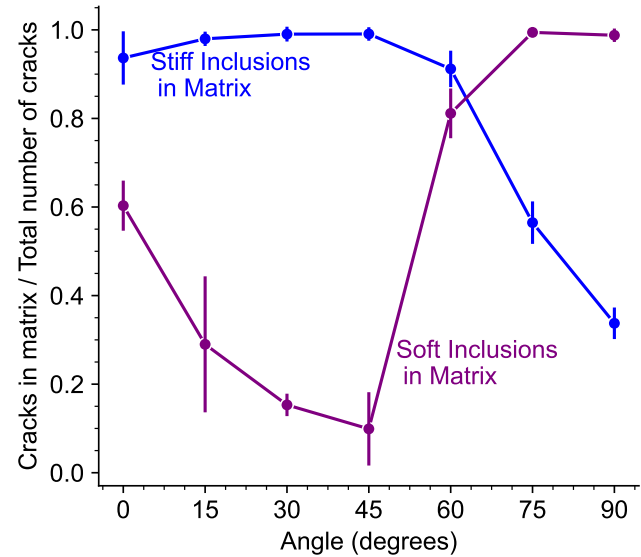


Fig. 18: Effect of angle of laminations with respect to the loading direction (Y-direction) on the crack distribution in the matrix material to the total number of cracks. Vertical bars at each point represent the standard deviation observed during different random realizations of the packing.

4 CONCLUSIONS

- The effect of the volume fraction of inclusions on mechanical properties is significant. As the volume of stiff material increases, mechanical properties such as Young's modulus, Poisson's ratio and UCS also increase. Conversely, an increase in the volume of soft material leads to a decline in these properties. This observation is consistent with experimental studies indicating the importance of inclusion volume in determining the mechanical behavior of composites.
- The number of laminations or their relative thickness has a minimal impact on the effective Young's modulus and Poisson's ratio, particularly when loading is applied parallel to the laminations. However, significant variations in Poisson's ratio are observed with loading perpendicular to laminations.
- The orientation of loading (parallel or perpendicular to laminations) significantly influences the initiation and propagation of cracks. Under parallel loading, stiff inclusions tend to initiate cracks within their region, while soft inclusions lead to cracks primar-

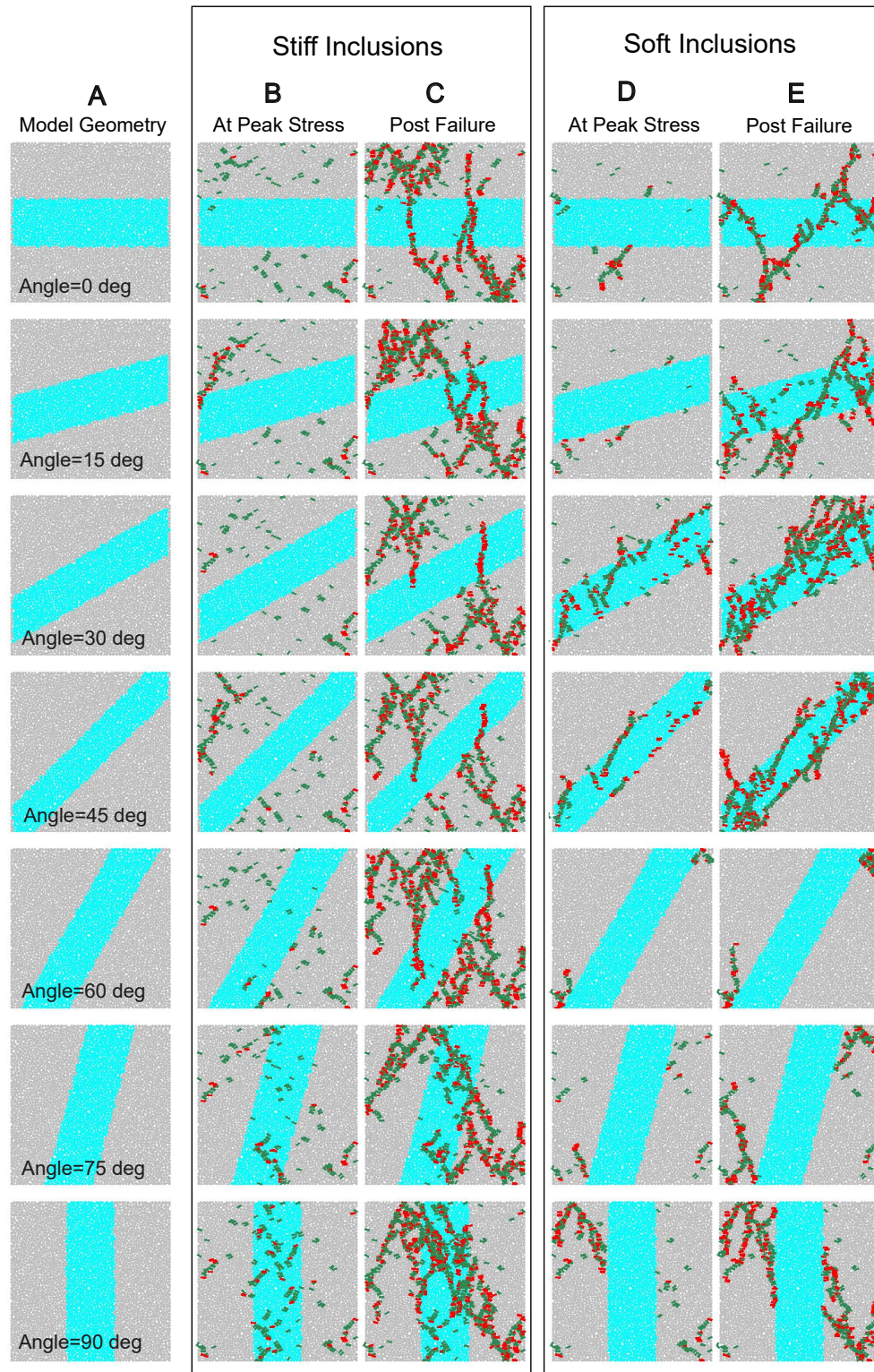


Fig. 19: Dynamic visualization of crack evolution during UCS test loading along Y-direction. Column (A) denotes model geometry (B) and (C) represent the crack distribution at the peak strength and post-failure of stiff inclusions. (D) and (E) show the crack distribution at the peak strength and post-failure stage. Green color and red color represent the shear and tensile cracks, respectively.

ily in the matrix. This indicates the role of material stiffness and loading direction in crack development,

with a negligible impact on the number of lamina-tions.

- The angle of laminations relative to the loading direction has a marked effect on Young's modulus, Poisson's ratio, and UCS, especially with soft inclusions. Young's modulus remains relatively constant up to 45 degrees and then increases, indicating angle-dependent anisotropic behavior. Moreover, the distribution and evolution of cracks are significantly influenced by the lamination angle, with a noticeable change in behavior at angles above and below 45 degrees.
- There is a distinct behavior observed between soft and stiff inclusions in terms of mechanical properties and crack evolution under various conditions (volume fraction, lamination number, and angle). This distinction highlights the complexity of composite material behavior, where both the mechanical properties of inclusions and their geometrical arrangement play critical roles.

REFERENCES

1. Aldred, R. (2021). Tutorial: Petrophysics of Thinly Bedded Formations. *Petrophysics - The SPWLA Journal of Formation Evaluation and Reservoir Description*, 62(04):335–352.
2. Aliabadian, Z., Zhao, G.-F., and Russell, A. R. (2019). Failure, crack initiation and the tensile strength of transversely isotropic rock using the brazilian test. *International Journal of Rock Mechanics and Mining Sciences*, 122:104073.
3. Amadei, B. (1996). Importance of anisotropy when estimating and measuring in situ stresses in rock. *International Journal of Rock Mechanics and Mining Sciences Geomechanics Abstracts*, 33(3):293–325.
4. Chenevert, M. E. and Gatlin, C. (1965). Mechanical anisotropies of laminated sedimentary rocks. *Society of petroleum engineers journal*, 5(01):67–77.
5. Cho, J.-W., Kim, H., Jeon, S., and Min, K.-B. (2012). Deformation and strength anisotropy of asan gneiss, boryeong shale, and yeoncheon schist. *International Journal of Rock Mechanics and Mining Sciences*, 50:158–169.
6. CUNDALL, P. A. and HART, R. D. (1993). 9 - numerical modeling of discontinua. In FAIRHURST, C., editor, *Analysis and Design Methods*, pages 231–243. Pergamon, Oxford.
7. Deere, D. U. and Miller, R. (1966). Engineering classification and index properties for intact rock. Technical report, Illinois Univ At Urbana Dept Of Civil Engineering.
8. Duan, K., Kwok, C., and Pierce, M. (2016). Discrete element method modeling of inherently anisotropic rocks under uniaxial compression loading. *International Journal for Numerical and Analytical Methods in Geomechanics*, 40(8):1150–1183.
9. Garcia, F. E. and Bray, J. D. (2018). Distinct element simulations of shear rupture in dilatant granular media. *International Journal of Geomechanics*, 18(9):04018111.
10. Heng, S., Guo, Y., Yang, C., Daemen, J. J., and Li, Z. (2015). Experimental and theoretical study of the anisotropic properties of shale. *International Journal of Rock Mechanics and Mining Sciences*, 74:58–68.
11. Heng, S., Li, X., Liu, X., and Chen, Y. (2020). Experimental study on the mechanical properties of bedding planes in shale. *Journal of Natural Gas Science and Engineering*, 76:103161.
12. Itasca Consulting Group, I. Timestep determination, PFC manual. Accessed: February 22, 2024.
13. Itasca Consulting Group, I. (2023). Flat-joint model â pfc 7.0 documentation. <https://docs.itascacg.com/pfc600/common/contactmodel/flatjoint/doc/manual/cmflatjoint.html#equation-ars>. (Accessed on 09/24/2023).
14. Li, J., Yu, Z., Zhou, Z., Wang, Y., and Li, J. (2021). Mechanical analysis and failure modes prediction of composite rock under uniaxial compression. *Scientific Reports*, 11(1):22826.
15. Moradian, Z., Einstein, H. H., and Ballivy, G. (2016). Detection of cracking levels in brittle rocks by parametric analysis of the acoustic emission signals. *Rock Mechanics and Rock Engineering*, 49:785–800.
16. Munjiza, A., Rougier, E., Knight, E. E., and Lei, Z. (2020). Discrete element and particle methods. *Encyclopedia of Continuum Mechanics*.
17. Ni, R., Ling, K., and Afari, S. (2023). Different ways to approach shale reservoirsâ co2 storage potential in america. *Heliyon*, 9(8):e18458.
18. Noori, M., Khanlari, G., Sarfarazi, V., Rafiei, B., Nejati, H. R., Imani, M., Schubert, W., and Jahanmiri, S. (2024). An experimental and numerical study of layered sandstone's anisotropic behaviour under compressive and tensile stress conditions. *Rock Mechanics and Rock Engineering*, 57(2):1451–1470.
19. O'Sullivan, C. (2011a). Introduction. In *Particulate Discrete Element Modelling, A Geomechanics Perspective*, page 32. CRC Press, 1st edition.
20. O'Sullivan, C. (2011b). Particle motion. In *Particulate Discrete Element Modelling, A Geomechanics Perspective*, page 46. CRC Press, 1st edition.
21. PFC7.0 Itasca Consulting Group, Inc (2021). PFC â Particle Flow Code, Ver. 7.0. Minneapolis: Itasca.
22. Plinninger, R. J. and Alber, M. (2015). Assessment of intact rock strength in anisotropic rock-theory, experiences and implications on site investigation. In *ISRM EUROCK*, pages ISRM-EUROCK. ISRM.
23. Pothana, P., Garcia, F. E., and Ling, K. (2024). Effective Elastic Properties and Micro-mechanical Damage Evolution of Composite Granular Rocks: Insights From Particulate Discrete Element Modelling. *Rock Mechanics and Rock Engineering*.
24. Russo, S., Qiu, J., Li, D., Li, X., and Zhou, Z. (2017). Dynamic fracturing behavior of layered rock with different inclination angles in shpb tests. *Shock and Vibration*, 2017:7687802.
25. Shi, Q. and Mishra, B. (2021). Discrete element modeling of delamination in laboratory scale laminated rock. *Mining, Metallurgy & Exploration*, 38(1):433–446.

26. Singer, V., Sautter, K. B., Larese, A., WÄEchner, R., and Bletzinger, K.-U. A partitioned material point method and discrete element method coupling scheme. 9(1):16.
27. Williams, P. F. (1990). Differentiated layering in metamorphic rocks. *Earth-Science Reviews*, 29(1):267–281.
28. Winhausen, L., Khaledi, K., Jalali, M., Bretthauer, M., and Amann, F. (2023). The anisotropic behavior of a clay shale: Strength, hydro-mechanical couplings and failure processes. *Journal of Geophysical Research: Solid Earth*, 128(11):e2023JB027382. e2023JB027382 2023JB027382.
29. Xu, H., Sun, X., Zhang, Y., Zhao, C., Miao, C., and Wang, D. (2023). Creep characteristics of layered rock masses after water absorption due to structural effects. *International Journal of Environmental Research and Public Health*, 20(5).
30. Yan, B., Wang, P., Ren, F., Guo, Q., and Cai, M. A review of mechanical properties and constitutive theory of rock mass anisotropy. 13(12):487.
31. Yang, S.-Q., Yin, P.-F., Huang, Y.-H., and Cheng, J.-L. (2019). Strength, deformability and x-ray micro-ct observations of transversely isotropic composite rock under different confining pressures. *Engineering Fracture Mechanics*, 214:1–20.